

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

Sebastian Bobadilla-Suarez
Independent Researcher
`sebastian.bobadilla.s@gmail.com`
ORCID: 0000-0002-5951-0772

19 September 2026

Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements and state the conditions that connect them: *coherence*, meaning collective phase order; *compatibility*, meaning that local descriptions agree where their domains overlap; and *self-representation*, meaning that the resulting state carries an accurate reconstruction of itself. Four results carry the argument. Installed energy bounds the mean coupling a declared actuator supplies, so a coherent stationary branch requires it to exceed $2D/\kappa$, with D the phase diffusion and κ set by the mode prices. Phase encoders shared on overlaps turn phase order into a bound on content disagreement, vacuous for a whole population and informative within a patch. Compatible local states glue to a unique global state. A contracting reconstruction that fixes that state makes it the unique fixed point, while accuracy across a family of states requires distinguishable encodings. Lean 4 checks these results and their composition, with the eight connecting hypotheses as explicit arguments and no axioms beyond its libraries'. Numerical controls bound what observation can decide: distinct mechanisms share stationary phase

distributions, finite ramps shift the apparent onset exponent, a winding field and a uniform one are stationary at one coupling with resultants spanning nearly the range of that observable, and a coupling rule descending a squared-drift objective holds coherence while losing the structure it was meant to acquire. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling. Testing it requires calibrated geometry, resolved phase dynamics, validated decoders and a spatial observable beside the resultant.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their interpretation requires evidence beyond the phase model.

Unity has been approached differently elsewhere: integrated information

theory treats integration and exclusion as constitutive (Tononi et al., 2016), global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene et al., 2014), and predictive-processing accounts derive both from a hierarchical generative model (Friston, 2010). Here the three requirements are kept apart rather than derived from one another, so that each carries its own assumptions and admits its own measurement; Section 8.4 returns to the comparison.

Consider reaching for a cup. A visual description of its location and a description of the hand’s relation to it can agree about where the cup is while contributing different contents. Synchronizing the two processes would leave disagreement about that shared quantity unresolved. An internal representation must additionally locate the perceiver within the represented situation. This example motivates the requirements; the mathematical constructions use spatial measures and fields in real normed vector spaces, whose identification with perceptual content is a further modelling task.

The mathematical contribution is to specify connections between these requirements and the assumptions each connection needs. A declared actuator’s stored energy bounds its mean coupling. Phase encoders shared on overlaps convert phase order into a bound on content disagreement. Compatible spatial measures have a unique global extension; approximate agreement controls a specified global selection. A contracting reconstruction makes that extension its unique fixed point only if it accurately reconstructs the extension. Accuracy across a family of states imposes a further condition on the

40 information retained by the encoding.

41 The empirical contribution is to identify what observations can decide.
42 Stationary phase distributions can be shared by stronger coupling, weaker
43 diffusion and a matched common drive. Finite ramps change the observable
44 onset. A winding field is stationary at the same coupling as a uniform one and
45 has the opposite resultant, so that observable does not determine the state.
46 Coherent adaptation can lose alignment with a designated environmental
47 template, and a compatibility estimator needs diagnostics that distinguish
48 content from decoder artifacts. These findings determine the measurements
49 needed to test the proposed connections.

50 Three of the ingredients are standard: the synchronization threshold
51 $K_c = 2D$ for the noisy mean-field model (Sakaguchi, 1988), the unique gluing
52 of compatible local sections, and the Banach fixed-point theorem (Banach,
53 1922). Four things here are not. The installed-energy condition bounds
54 mean coupling by the stored energy of a declared actuator, which makes a
55 resource condition necessary for coherence on the stationary branch. The
56 coherence-to-content bound converts phase order into a statement about de-
57 scriptions rather than about timing, and its patch-local form is what keeps
58 it from being vacuous in a large population. A stability criterion on the
59 reweighted coupling kernel, together with an existence condition on an am-
60 plitude layer, settles when a low resultant reports weak coupling and when
61 it reports a winding field. And the composition is machine-checked together
62 with a witness discharging all eight hypotheses at once, which is what makes

63 their conjunction non-empty rather than a list no object satisfies.

64 Granting those premises, collected in Table 1, the account runs as follows.
65 Extracellular contraction on awakening raises endogenous field amplitude
66 and with it the effective coupling between cortical oscillators. Once that
67 coupling passes $2D$, a coherent phase branch exists, and from an initial spread
68 within a quarter turn the population settles onto it; other initial data leave
69 windings stationary at the same coupling. Encoders shared where cortical
70 territories overlap turn that phase order into agreement about the quantities
71 the territories describe in common, so the local descriptions glue: there is
72 one state of which each is a restriction. Part of that state encodes the state
73 itself, and a readout of that part reconstructs it accurately enough, and
74 sensitively enough across the states that matter, to make it the unique state
75 the system models itself as being. On this account the glued state is what
76 makes distributed activity one situation, and its accurate self-reconstruction
77 is what gives that situation a perspective. Every step of it is conditional.

78 Sections 2–4 develop the three requirements together (Fig. 1). Section 5
79 composes them, stating the thermodynamic constraints and all eight con-
80 necting hypotheses per branch; its supporting models of feedback, learning
81 and resource supply have distinct state spaces and assumptions. Section 6
82 then examines what the observables can establish, before Section 7 devel-
83 ops the cortical hypothesis and an awakening experiment. The hypothesis is
84 that extracellular contraction increases field-mediated coupling; its direction,
85 rate and relation to recovery require independent calibration. Supplemen-

tal Material organizes the constructions and controls behind the claims they support.

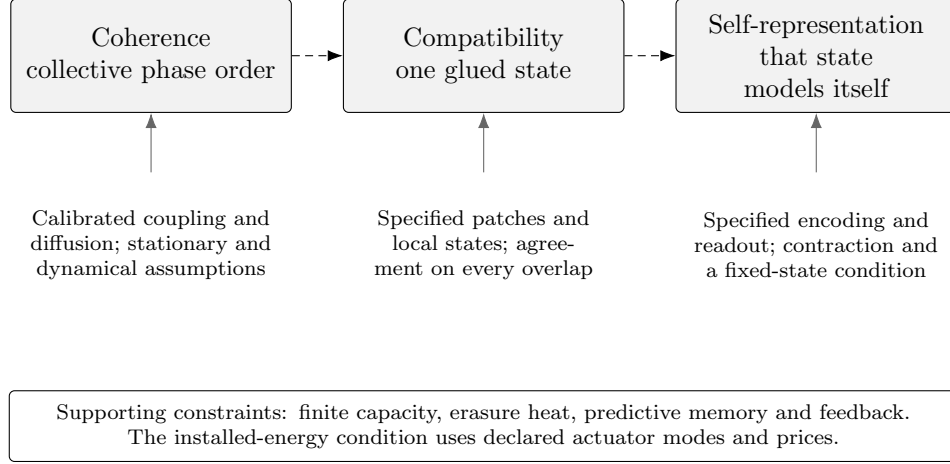


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

88 2 Coherence: physical coupling and its re- 89 source condition

90 2.1 Dynamics and the stationary threshold

91 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

92 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
93 K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field
94 coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct
95 parameters. The stationary calculation below uses identical frequencies and
96 this scalar coupling. Applying it to a spatial kernel requires an aggregation
97 rule and a check of the resulting approximation.

98 For identical frequencies in their rotating frame, the positive normalized
99 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

100 where I_n are modified Bessel functions and ψ is the collective phase. The for-
101 mal derivation uses the absence of stationary current for a periodic gradient
102 drift to obtain this density, then verifies normalization and self-consistency.
103 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988). Above

threshold, exactly one positive solution coexists with zero and increases
strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

Figure 2 shows this stationary branch.

Linearization about the uniform density gives rate $K/2 - D$ for the first harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists exactly above threshold. About the coherent density, the quadratic form is strictly dissipative transverse to collective rotation, whose rate is zero. These linear estimates concern the limiting equation. Relating them to finite stochastic trajectories requires a dynamical mean-field limit and branch selection; exchange symmetry alone does not establish the needed independence (Sznitman, 1991).

2.2 Installed energy bounds mean coupling

A spatial model replaces the finite sum by integration against a kernel. One declared implementation carries response modes with continuous profiles ϕ_i , installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

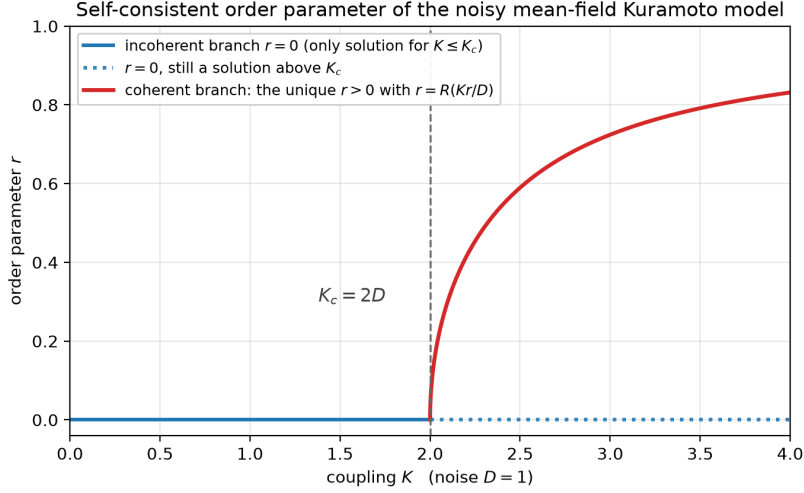


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

119 The same configuration sets the reciprocal kernel and stored energy. An
 120 executed transition changes the kernel through its changed occupancies and
 121 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The
 122 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 123 pair reconstruction on refining spatial partitions converges to its product-
 124 measure integral under the specified compactness and sampling assumptions;
 125 the refinement index is independent of the number of control updates.

126 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 127 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's

128 integral gives

$$\begin{aligned}
 U_{\text{inst}} &= \sum_z p(z) U(z), \\
 K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}.
 \end{aligned} \tag{5}$$

129 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 130 coherent stationary branch and any growing Fourier mode about the uniform
 131 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 132 is necessary: energy above it need not produce sufficient coupling because
 133 the bound can be loose.

134 The quantity constrained is installed energy. Dissipated heat does not
 135 determine it, and κ is supplied by the mode profiles, prices and substrate
 136 normalization. Applying the scalar threshold to heterogeneous or signed spa-
 137 tial dynamics requires a justified reduction. Identifying the installed kernel
 138 with a cortical field is a separate physical commitment.

139 **3 Compatibility: from coordinated activity** 140 **to a global description**

141 **3.1 Why common timing is insufficient**

142 For overlapping patches U and V , compatibility means that restricting either
 143 local description to $U \cap V$ gives the same object. Descriptions can agree on

144 shared quantities while contributing different information elsewhere. Com-
 145 patible sections of the specified sheaf have a unique global extension. On
 146 a finite cover, local finite spatial measures agreeing as measures on every
 147 overlap also have a unique finite-measure extension whose sheaf image is the
 148 glued section. No mass-one normalization is imposed.

149 In the cup example, visual and reaching descriptions must express a
 150 shared quantity, such as cup location, in common coordinates. Restriction
 151 retains that quantity while dropping details particular to each description.
 152 Agreement permits surface appearance and hand configuration to contribute
 153 to one situation. The mathematical counterexample assigns the same phase
 154 to every patch while two local mass profiles disagree on a shared site. Those
 155 profiles have no common extension. Because they are chosen independently
 156 of phase, synchrony places no constraint on their disagreement.

157 This distinction limits the biological interpretation. Receptive territories
 158 and content domains must be related to spatial patches, and the descriptions
 159 must support the required extension. Matching overlap marginals of de-
 160 coded probability laws need not determine a unique global joint distribution
 161 (Abramsky & Brandenburger, 2011).

162 **3.2 When coherence does constrain content**

163 A declared relation between phase and content gives a positive result. Let
 164 contents take values in a real normed vector space E . Patch i reads an
 165 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point

166 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 167 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 168 sites; their equality on overlaps is a supplied relation between descriptions of
 169 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 170 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 171 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

172 Here discrepancy is the norm of the difference. At $r = 1$, the observa-
 173 tions agree exactly. The factor N makes this estimate potentially weak
 174 in large populations; it supplies a pointwise guarantee from a population
 175 average. The population in it need not be the whole array. A patch car-
 176 ries an order parameter of its own, and the same bound read there car-
 177 ries that patch's resultant and that patch's site count (`chord_le_of_patch_`
 178 `coherence`, `compatible_of_patch_coherence`), so a large population ob-
 179 served through small patches is constrained by the small count. A shared
 180 scalar encoder is a special case. Dropping the overlap relation permits dis-
 181 agreement even at perfect locking. A common nonexpansive linear projection
 182 preserves agreement, but agreement of a projection does not establish agree-
 183 ment of the full vector.

184 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 185 ηo_i . Content and observation discrepancies bounded by ε and δ give a next

186 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 187 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

188 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 189 duce an initial mismatch. Approximate observations leave a nondecaying
 190 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 191 update to the phase process.

192 **3.3 What approximate agreement determines**

193 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 194 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 195 Weights that vanish outside their patches and sum to one at each site select
 196 a global profile by weighted averaging. The selection lies within ε of each
 197 patch there, and any two such selections differ by at most ε , independently
 198 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 199 $L\varepsilon$ across selections.

200 At zero tolerance, the selection is the unique exact extension. At every
 201 positive tolerance there are families with no exact extension; selection then
 202 gives an approximation whose dependence on weights is bounded. Supple-
 203 mental Material gives a three-site witness attaining the bound. The mass-
 204 profile construction uses the scalar case of the content dynamics and addi-
 205 tionally requires nonnegativity of the initial profiles and observations. The

206 general vector result supplies no identification with the spatial-measure sheaf.

207 These constructions concern declared objects: a specified cover, contents
208 valued in a normed space, and encoders related on overlaps. They select
209 neither the neural cover nor the decoder. Identifying their contents with
210 perceptual content, and establishing that the identification bears on experi-
211 ence, is a further modelling task; Section 6 sets out what a measurement of
212 it would have to establish.

213 **4 Self-representation: a state that models it-** 214 **self**

215 **4.1 The reflexive construction**

216 Compatibility gives a global state. Reflexivity relates it to an encoding within
217 that state. Equip global sections with a complete metric. If the additional
218 restriction-resonance condition identifies encoding with restriction to a region
219 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \tag{8}$$

220 The region and readout are fixed across the state space. If F contracts and
221 fixes the section s_* formed from the local descriptions, Banach's theorem
222 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89

223 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
 224 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
 225 one for $\tau > 0$. Constructing a map with that factor remains a modelling
 226 obligation.

227 In the cup example, the global state includes the represented scene and the
 228 perceiver's relation to it. A local encoding must retain enough information
 229 for the readout to recover that modelled state; cup position alone would be
 230 insufficient if hand configuration and viewpoint could not be reconstructed.

231 For the same map with Lipschitz factor $q < 1$, approximate reconstruction
 232 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (9)$$

233 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 234 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 235 weakens.

236 4.2 Reconstruction across a family of states

237 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 238 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 239 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

240 Thus M relevant states pairwise more than 2ε apart require at least M
 241 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 242 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 243 existing self-prediction map.

244 This is an additional accuracy criterion on a family, with no requirement
 245 for multiple exact fixed points. A blind encoder or constant readout fails it
 246 for separated states despite having a unique fixed point.

247 **4.3 Agreement after a change takes communication**

248 Snapshot compatibility does not specify how quickly descriptions respond
 249 when a shared quantity changes. In a synchronous network whose updates
 250 read only the site’s own state and declared neighbours’ messages, two initial
 251 configurations agreeing on a site’s causal past through T rounds produce the
 252 same state there at round T . If two fresh interventions outside that past
 253 require reports separated by more than 2ε , no report from that site can be
 254 within ε of both. This follows from the reconstruction bound in Section 4.2.

255 A three-site delay line witnesses failure after one round and exact report-
 256 ing after two; cutting the path prevents delivery at every round. A report
 257 may still happen to be correct about one fixed world. The bound concerns
 258 responsiveness across interventions.

259 Throughout this section *prediction* names a reconstruction map; reading
 260 it as temporal neural prediction would require dynamics. The relevant family
 261 of states, the metric, the tolerance, the communication graph and the dead-

line are all declared inputs, as are the represented variables and the fidelity counted as accurate. Reading the resulting code count as physical memory capacity requires a hardware model, and a readout that varies with the state requires a further one.

5 Thermodynamics and composition

The phase and representation results concern coupling, content disagreement and reconstruction error. Thermodynamics constrains the physical operations that could implement them. Finite-register capacity obeys $H(p) \leq \log |X|$; erasure heat depends on specified system–bath assumptions (Landauer, 1961; Norton, 2005; Shenker, 2000). A separate boundary result concerns a field reaching disconnected vacuum components (Kibble, 1976). The composition connects these objects through the explicit hypotheses in Table 1.

5.1 Feedback changes the information constraint

For passive memory X_t driven by an environmental signal S_t , the predictive bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still et al., 2012). Zero retained information also satisfies it, so a task and a

280 learning rule require additional specification. Relating this bound to Lan-
 281 dauer heat requires a common thermal scale and budget, and memory waste
 282 bounded by erased entropy.

283 An acting agent changes the signal it will receive. Specify policy, world,
 284 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
 I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
 &\quad - I(X_t; S_{t+1} \mid S_t).
 \end{aligned}
 \tag{12}$$

285 The second conditional term vanishes in the passive model. With feedback,
 286 the signed difference can be negative as action creates correlations with the
 287 future. A causal reading requires an environmental state sufficient to describe
 288 the relevant dynamics.

289 For a controlled substep, the active thermodynamic bound instead uses
 290 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
 291 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
 292 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
 293 2013). Every forward path of positive probability must have a positive-
 294 probability reverse for this finite identity. A missing reverse gives infinite
 295 divergence. The supplement develops the support conditions and composes
 296 actuation with memory updates on their shared intermediate law.

297 **5.2 Learning and repeated operation need explicit re-** 298 **sources**

299 The finite examples distinguish selecting a feasible policy, relaxing a register
300 toward an encoded reward, and learning an unknown task from observa-
301 tions. The observational learner acquires correlation with an environmental
302 parameter while its action marginal stays uniform. A continuing model exe-
303 cutes actuation, observation-driven learning and reset in sequence, preserving
304 above-chance performance at a positive work cost per cycle.

305 A phase-derived sensor makes the informational distinction concrete: an
306 informative readout and a blind control have the same phase order but dif-
307 ferent learning performance. Attaching the learner’s register to the actuator
308 of Section 2.2 ties reward, installed kernel and work to its own executed law.
309 The learning witness rises above the necessary installed-energy bound while
310 its matched control remains on the exclusion line. These finite constructions,
311 detailed in Supplemental Material, leave the evolving feedback between phase
312 dynamics, installation and sensing to be specified.

313 For that model, common energy E , thermal scale θ , local detailed balance
314 and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (13)$$

315 where expected cumulative work stays within the supplied allowance B at
316 every prefix. This bounds expectations. Separate constructions track a store

on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles, while continuing operation requires replenishment. Register–bath witnesses also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 2.2 requires physical relations that these separate models do not supply automatically.

5.3 What the composition establishes

The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named substep at the register’s thermal scale satisfies

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

with B the register’s heat. The upper allocation follows from a ledger whose operations deliver that total heat to one reservoir and whose other operations do not increase joint entropy. Those ledger assumptions are physical inputs. Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12	Physical	A finite-capacity system realizes a field
<i>capacity and</i>	premise	leaving its vacuum manifold. Capacity and
<i>boundary</i>		boundary are independent roots.
E23	Bridge	The boundary is associated with a finite
<i>boundary to</i>	assumption	register whose update leaves states
<i>register</i>		unreachable.
E34 /	Bridge	Passive: predictive memory at the register’s
E34Active	assumption	thermal scale and heat budget, with
<i>register to</i>		nonzero waste. Active: positive paths, local
<i>dissipation</i>		detailed balance and allocation of register
		heat to the named substep; the allocation
		uses the ledger identity and other
		operations’ compressiveness.
E45 /	Modelling	Convergent spatial coupling energies.
E45Active	assumption	Declared mode hardware supplies a kernel
<i>couplings to</i>		limit for either branch. A learning witness
<i>continuum</i>		computes installation and work from its
		own executed law. Refinement and physical
		identifications remain inputs.
E56	Physical	A specified physical field realizes the
<i>field</i>	commitment	limiting coupling and satisfies the kernel
<i>identification</i>		conditions.
E67	Physical	That field operates above the
<i>above</i>	commitment	synchronization threshold.
<i>threshold</i>		
E78	Bridge	Coherence relaxes the supplied cover at
<i>coherence to</i>	assumption	coupling at least the named mean-field
<i>cover</i>		value. Shared phase encoders bound
		vector-content mismatch; the exact cover
		and timely responsiveness remain separate
		requirements.

Table 1 (continued)

Connection	Status	Required assumption
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 2.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and readout. They conclude **UnifiedSelf**: the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (8).

Each of the eight edges is supplied rather than derived from the one above it. E12 is equivalent to its conclusion, so capacity and boundary are independent roots rather than a step from one to the other; both E45 discharges take their convergence from refinement data and not from the thermodynamic bound; E78 states that coherence relaxes the supplied cover, and the coherence-to-content estimate of Eq. (6) bounds disagreement only for encoders already shared on overlaps. The inferential content of the composition is its last step, where contraction and a fixed glued section give uniqueness. What composing them establishes is that the premises are consistent, named

350 in one place and machine-checked together, and not that coherence follows
351 from installed energy or self-representation from coherence.

352 Joint witnesses combine a register, boundary, refining mesh and small
353 spatial cover; active witnesses add controlled thermal updates and a priced
354 installation. They establish satisfiability under their stated identifications.
355 Mesh convergence follows from refinement data, and the witnesses establish
356 no common cortical mechanism. Supplemental Material maps the claims to
357 their Lean declarations and records the counterexamples. The development
358 declares no additional axioms beyond its libraries' foundational axioms.

359 **6 What the observations establish**

360 The composition holds under its premises. Whether observation can establish
361 those premises is a separate question, and the three requirements it connects
362 concern different objects: phase dynamics, local descriptions and internal
363 encodings. Their observables must retain these distinctions. The following
364 controls show which mechanisms remain indistinguishable, which temporal
365 approximations fail, which states one observable cannot separate, and what
366 independent content measurements must establish.

367 6.1 Stationary distributions leave the mechanism un- 368 determined

369 Equation (2) gives a concentration–coherence relation with no fitted shape
370 parameter. Estimate r from circular resultant length and a from density
371 shape: inverting R from the measured r would make agreement automatic.
372 Even independently estimated agreement identifies a statistical family shared
373 by different mechanisms.

374 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
375 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
376 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
377 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
378 and diffusion D_0 have that same density. Their stationary distributions co-
379 incide window by window despite different causes.

380 Resolved dynamics supply additional information. In the ideal model with
381 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
382 increment means constrain drift. Separating endogenous coupling from a
383 matched common drive additionally requires a measured drive or controlled
384 perturbation; identifying electromagnetic coupling requires evidence specific
385 to that interaction. Supplemental Material gives the equivalence construction
386 and its numerical check.

387 **6.2 Finite ramps change the observable onset**

388 The square-root law in Eq. (3) describes stationary solutions. A changing
389 coupling requires time for the phase distribution to approach those solutions.
390 In a deterministic mean-field control that crosses the threshold symmetrically,
391 none of the scanned rates meets the declared tracking tolerance: slowing
392 the crossing does not remove the terminal residual under that initialization.
393 A slow control variable therefore supplies no general guarantee of branch
394 tracking at onset. Supplemental Material gives the residual, rate range and
395 admissible rates away from threshold.

396 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
397 at four coupling speeds. Escape is the first of three consecutive saved samples
398 at $r \geq 0.2$ after threshold. Faster ramps delay escape (Fig. 3); only 5/32
399 replicas escape within the fastest window. The three slower ramps have
400 mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted onset exponents
401 differ from those of the exact stationary branch evaluated on the same grids.
402 At the fastest speed the ensemble mean never reaches the fit window, so its
403 fitted slope does not estimate stationary onset.

404 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
405 makes the apparent exponent depend on ramp speed and fitting window.
406 The specified deviation-based critical ramp speed is not bracketed by this
407 sweep. With fluctuating coupling, the relevant stationary comparison also
408 depends on rate: the mean coupling approximates order in the tested fast
409 regime, while averaging stationary order over coupling works for the tested

slow drives confined to one side of threshold. Slow crossing drives can satisfy
neither approximation. These are finite-horizon observations, with controls
and parameters in Supplemental Material.

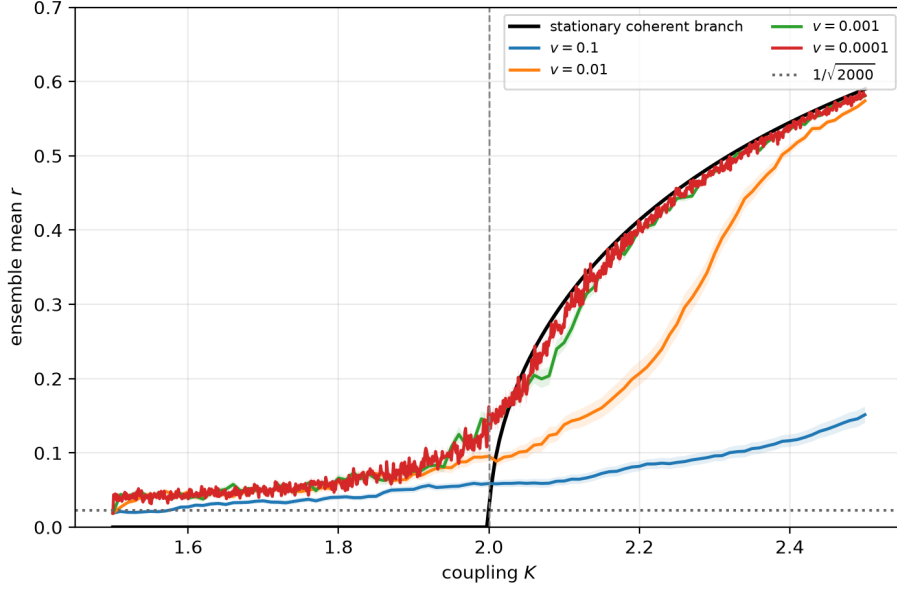


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

Additional controls address integration bias, population size and spatial
or signed coupling. The refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 $[1.958, 2.009]$ contains 2. These are finite numerical regimes, with full pa-
rameters and controls in Supplemental Material; they supply no cortical cal-
ibration.

418 6.3 A winding state leaves the order parameter am- 419 biguous

420 The order parameter r is indifferent to where a phase sits in space. A field
421 winding once across a periodic sheet therefore has $r = 0$ while every patch
422 of it stays locked. Such a field is not an artificial construction here. On a
423 torus a winding is stationary for any isotropic symmetric kernel at identical
424 frequencies: the coupling term is odd in the separation against a kernel even
425 in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves
426 this for any finite periodic array, this sheet included. A winding satisfies the
427 positivity and identical-frequency hypotheses. It violates only the quarter-
428 turn initial spread that the convergence result additionally requires before
429 placing the limit at the uniform state.

430 On the sheet of Section 2's spatial control, a winding survives below
431 an interpolated decay length of 0.1871 mm and unwinds above it. That
432 boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same sheet at the
433 same coupling, diffusion and kernel reaches $r = 0.9657$ from a uniform start
434 and $r = 0.0083$ from a winding one. Both are stationary, with patch-local
435 order 0.9015 and no winding defect. At fixed K the observable therefore takes
436 two values spanning nearly its whole range. Coupling does not determine the
437 resultant, and a low resultant does not by itself distinguish weak coupling
438 from a winding field.

439 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-

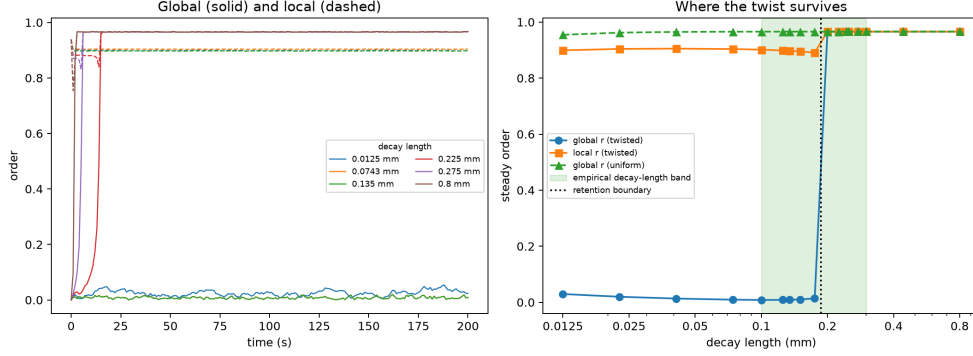


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

440 trol carries, the boundary moves to 0.1421 mm. The same comparison at
 441 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.
 442 Heterogeneity shortens the range that sustains a wave without removing it.

443 Which windings persist has a criterion behind it. Linearizing the phase
 444 model about a winding returns a coupling of the same translation-invariant
 445 form, each separation reweighted by the cosine of the phase the winding
 446 advances across it (`hasDerivAt_charJacobian`). That reweighted kernel is
 447 again even in the separation, so the same characters diagonalize it and ev-
 448 ery mode’s rate is one of its eigenvalues. Where the winding advances by
 449 at most a quarter turn across every separation the coupling reaches, the

450 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
 451 `stable_of_quarter_turn`); where some rate is positive, that mode grows
 452 without bound (`charJacobian_growing_mode`). Moving coupling mass to
 453 longer separations is what carries a phase advance past a quarter turn, and
 454 that is the mechanism the decay-length sweep varies. The criterion bounds
 455 rates rather than trajectories: asymptotic stability is not proved, so the
 456 states that persist occupy a narrower range than it admits, and the bound-
 457 ary quoted above is measured and not derived from it.

458 The same reweighted kernel measures the potential's curvature. A wind-
 459 ing has $r = 0$ where a global minimizer of the Lyapunov potential has $r = 1$,
 460 so it is not one (`char_not_potential_min`): the quarter-turn condition on ini-
 461 tial data does not merely decline to cover a winding, it must exclude it. Along
 462 a mode, the potential's second variation is that kernel's quadratic form, so a
 463 growing mode is a direction of falling potential and a winding carrying one
 464 is not a local minimum either (`char_not_local_min_of_charLambda_pos`).

465 An amplitude layer decides which of these states exist, and leaves what
 466 the resultant reports unchanged. A phase state carries no amplitude. Ev-
 467 ery oscillator sits on the unit circle, so a field with an amplitude node has
 468 no representation in these dynamics. The real Stuart–Landau extension
 469 supplies one on the same coupling, and leaves intact the gradient struc-
 470 ture the convergence results rest on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i =$
 471 $(\mu - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its phase equation is the Kuramoto equation
 472 with each coupling weighted by a ratio of amplitudes (`amplitude_phase_is_`

473 kuramoto). The phase model is this one with the amplitude ratios set to
474 one, rather than a separate object.

475 What the added degree of freedom decides is which windings exist at all.
476 On a periodic array of any dimension, this sheet included, a winding requires
477 $\mu + \lambda > 0$. Here $\lambda = \sum_d f_d(\cos \phi_d - 1)$ is the eigenvalue the kernel supplies,
478 summed over separations, with ϕ_d the phase the winding advances across d
479 (exists_charState_iff; on the ring $\phi_d = 2\pi qd/n$). For the nonnegative
480 kernels used here $\lambda \leq 0$, and moving coupling mass to longer separations
481 lowers it further. A broad enough kernel therefore removes the winding out-
482 right. The sweep above reaches that same boundary by a different route:
483 the phase model retains every winding at every coupling, and loses this one
484 under integration instead. The two layers bound the same regime from op-
485 posite sides. A phase recording does not separate a state the amplitude layer
486 excludes from one the dynamics fails to sustain.

487 These states are windings and not spirals, and on the sheet that is a
488 theorem rather than a shortage of dimensions. A cortical phase singularity
489 is a point at which the amplitude vanishes and around which the phase turns
490 once, read in practice as the circulation around a small plaquette. The states
491 above have equal modulus at every site, so that circulation is exactly zero
492 on every plaquette (plaquetteCirculation_eq_zero_of_winding). Nor can
493 the layer host a single defect. Over a periodic sheet the circulations of *any*
494 phase field sum to zero, so defects arrive in cancelling pairs (no_isolated_
495 defect). The sweep's near-zero measured defect density agrees with the first

496 of these and says nothing beyond it.

497 Two limits bound the layer for a reader of the cortical literature. Relax-
498 ational dynamics has no propagation speed, which is the quantity a cortical
499 wave measurement leads with. And $|z|$ is a normal-form radius, which a mea-
500 sured envelope is not; this work identifies the two nowhere. Supplemental
501 Material gives the sweep, its uniform control, the amplitude layer and the
502 standing wave it admits inside the band, together with the remaining limits
503 of both.

504 A vanishing resultant is the absence of global agreement and not of agree-
505 ment. Eq. (6) read on a nearest-neighbour patch of a winding gives a dis-
506 crepancy bound of $2\sqrt{2}L|\sin\phi|$, with ϕ the phase advanced across the patch,
507 which falls to zero as the same winding is spread over more sites (`chord_le-`
508 `of_char_patch`), while the global form of the same bound is vacuous at $r = 0$.
509 Encoders shared on overlaps therefore still bound content disagreement in-
510 side a patch of a state that no global observable calls coherent. The winding
511 number itself stays unconnected to content.

512 This qualifies what a phase trace can establish. An onset criterion stated
513 in r alone registers a change of basin as readily as a change of coupling.
514 The protocol of Section 7.3 therefore needs a spatial observable beside the
515 resultant.

516 **6.4 Coherent adaptation can lose environmental align-** 517 **ment**

518 An adaptive coupling rule might supply representational structure, but its
519 objective must reward the relevant agreement. For symmetric coupling, pair-
520 wise sine terms cancel in the population mean drift. Minimizing squared drift
521 at that fixed mean therefore rewards equalizing drift across nodes. In the
522 frequency-cluster control, this favors coupling between unlike-frequency clus-
523 ters, whereas the diagnostic environmental template rewards coupling within
524 clusters.

525 The formalization proves descent of the squared-drift objective for a spec-
526 ified gradient flow at fixed phases. Its functional differs from thermodynamic
527 entropy production: at a coherent stationary density the probability current
528 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
529 rial gives the operator, its gradient and the conditions of the descent result.

530 The finite control co-evolves noisy phases and symmetric, nonnegative
531 couplings at fixed total resource. Its template is a diagnostic covariance of
532 latent signals, never a training input. Across three seeds, second-half order
533 remains between 0.9167 and 0.9222, while the within-cluster to between-
534 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5).
535 Random-update, permuted-gradient and frozen-kernel controls are reported
536 in Supplemental Material.

537 The immediate post-update objective reduction disappears when aver-

aged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

6.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on constructed data separates these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial; the compatible construction scores 0.476 on an interval containing one half. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

Phase agreement cannot replace that measurement. On constructed con-

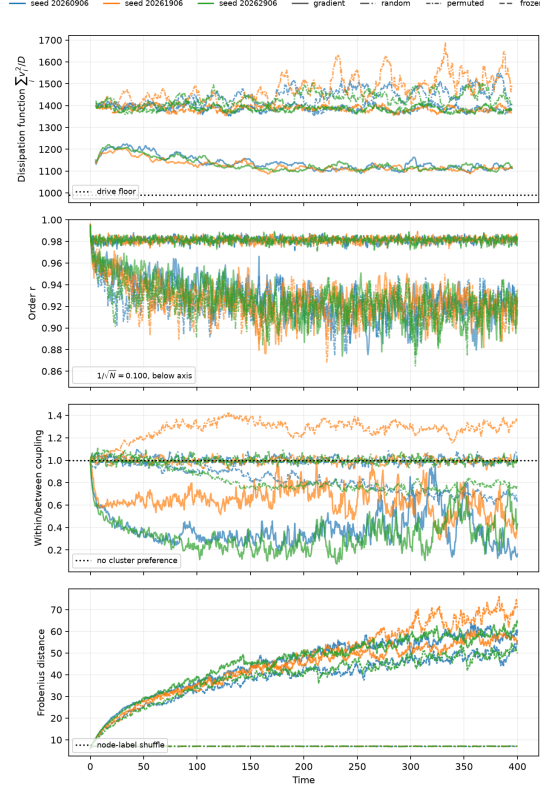


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

560 tent, phase carries a share 1.000 of the content entropy when content is built
561 from it and 0.000 when it is not. A phase-derived statistic reports com-
562 patibility on phase-locked patches whose local descriptions disagree. The
563 implemented content statistic detects their incompatibility. These controls
564 validate a measurement on known constructions; a neural content model and
565 decoder remain to be established.

566 Choosing the cover also chooses a candidate subject: trivial descriptions
567 can agree while carrying little content, and a partition can conceal disagree-
568 ment. The useful choice must therefore be constrained by what neural ac-
569 tivity encodes. The exact composition assumes compatibility of its supplied
570 cover. The content mechanisms above identify additional hypotheses under
571 which agreement can arise; they leave the empirical choice of content and
572 the connection to that cover explicit.

573 **6.6 What an informative phase-density measurement** 574 **requires**

575 The concentration range and the dependence structure decide whether the
576 concentration–coherence relation can be tested at all. Under the specified
577 log-density estimator and bias-floor criterion, discrimination at the observed
578 concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned
579 designs, rising to 31000 when sites share a mean in groups of 10. These
580 counts describe that estimator and dependence model; the intended recording

581 requires its own calibration. Supplemental Material reports the bin-count
582 controls and the full dependence-aware design.

583 That ceiling comes from exercising the estimators on public propofol-
584 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms windows.
585 Bipolar and common-reference montages produce different residuals. The
586 bipolar results are compatible with the Bessel relation over a concentration
587 range too narrow to distinguish it from its linear approximation at the ob-
588 served scatter, which is the range the design scan takes as given. Temporal
589 pooling changes the estimand, and sedation recordings leave the awakening
590 hypothesis untested.

591 An informative dataset therefore needs sufficient concentration range or
592 precision, calibrated estimators and an observation model for dependence,
593 source mixing and reference choice. Public mouse calcium imaging provides
594 thousands of spatially located neurons across waking, sleep and anesthe-
595 sia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-
596 estimation band used here. Site count, state coverage and bandwidth are
597 separate requirements.

598 **7 The cortical hypothesis and an awakening** 599 **test**

600 The cortical proposal identifies a physical source of coupling and asks whether
601 its changes help explain recovery. The tests in Section 6 determine what

602 observations must distinguish before a phase trace can support this identifi-
603 cation.

604 **7.1 Physical identification and calibration**

605 The cortical hypothesis identifies part of effective coupling with endogenous
606 extracellular electromagnetic interactions. Fields can modulate spike timing
607 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
608 propose coordination of distributed neural information (Pinotsis & Miller,
609 2022; McFadden, 2020). Phase organization under propofol motivates study-
610 ing this regime (Bastos et al., 2021), but does not measure coupling.

611 The formal field predicate requires a jointly continuous kernel, a normal-
612 ized substrate, coupling equal to the kernel average, and no single site pro-
613 viding more than half that strength. Toy constructions satisfy it. A cortical
614 realization must specify the scale at which the kernel and its reduction ap-
615 ply: calcium imaging reports different state-dependent modularity patterns
616 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

617 The spatial reduction is also testable. A row-sum aggregation at zero
618 diagonal tracks the uniform-coupling control over the tested spatial decay
619 range, while frequency heterogeneity displaces the operational threshold. The
620 fit window biases both estimates relative to $2D$; agreement with the control
621 is therefore the supported conclusion. Supplemental Material reports the
622 reduction, its numerical error and the distinct assumptions needed for the
623 kernel-average identification in Eq. (5).

624 The field commitment requires physical evidence beyond a choice of math-
625 ematical representation. A finite matrix spread over positive-mass cells pre-
626 serves its total coupling and phase correlation exactly. Point-supported
627 vanishing under an atomless measure and a disadvantage from excluding
628 a better-correlated pair concern specified support constraints. Neither com-
629 parison supplies energy prices or a biological–digital distinction.

630 A spike-time shift alone supplies no coupling rate. The dimensionless
631 product of population size N , timing sensitivity s , field amplitude E and
632 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (15)$$

633 This rate describes how induced timing shifts accumulate into drift. Nei-
634 ther it nor the installation factor κ follows from a heat bound. Testing the
635 physical threshold requires independent estimates of the interaction kernel,
636 its aggregation and diffusion in the same cortical regime.

637 **7.2 Extracellular geometry and recovery**

638 The proposed control variable is extracellular geometry. Extracellular vol-
639 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
640 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
641 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
642 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,

an effective-medium description predicts that a larger extracellular fraction raises conductivity and reduces field amplitude.

The coupling-gated hypothesis consequently predicts that extracellular contraction on awakening raises E and, holding the other factors in Eq. (15) fixed, raises effective coupling. This argument supplies a direction; the geometry–coupling relation, its magnitude and its role in human recovery require measurement.

For a smooth crossing at nonzero speed, evaluating the stationary branch along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

This follows from Eq. (3) at fixed D and concerns the stationary branch evaluated at $K(t)$. Reading it as observed recovery requires an independently justified tracking approximation. The crossing controls in Section 6.2 leave that approximation unsupported at onset, even though a separation of geometry and phase time scales can support it away from threshold. The awakening hypothesis therefore needs a calibrated dynamical prediction for an operational onset above the finite-size fluctuation floor. Crossing K_c alone does not supply a recovery latency.

7.3 A protocol for the awakening experiment

The physical test has six requirements:

- 662 **1. Coupling and diffusion in the same units.** Estimate them from re-
663 solved phase dynamics, separating heterogeneity and observation noise.
664 Measure the drive or perturb the system to distinguish endogenous cou-
665 pling.
- 666 **2. Geometry calibrated to coupling.** Validate the geometry observ-
667 able and its relation to effective coupling, with uncertainty. The con-
668 ductivity argument supplies only the expected direction.
- 669 **3. Calibrated phase measurements.** Estimate r by resultant length
670 and a by unweighted regression of log density on centered cosine.
671 Choose site count and binning from a dependence-aware calibration
672 over the intended concentration range. The tested designs require
673 thousands of sites at the observed concentrations; temporal pooling
674 changes the spatial estimand.
- 675 **4. An independent recovery endpoint.** Measure behavioral recovery
676 separately from phase organization.
- 677 **5. A predeclared dynamical onset criterion.** Set a coherence thresh-
678 old above the finite-size floor and evaluate its latency under the mea-
679 sured coupling trajectory, including ramp speed or fluctuations. Record
680 a spatial observable beside the resultant, since a winding field reports
681 a low resultant at coupling that sustains local order (Section 6.3).
- 682 **6. Matched model comparison.** Compare coupling increase, diffusion

683 change, common drive and alternative recovery models under the same
684 observation and filtering assumptions.

685 Failure of the concentration–coherence relation over an informative range
686 constrains the density model. Failure of the calibrated geometry–coupling
687 association constrains the physical gate. Better prediction of independent
688 recovery by a competing mechanism counts against its proposed role. A
689 favorable outcome still leaves the content of coordinated activity to be ex-
690 plained.

691 8 Discussion

692 The account set out in Section 1 is conditional at every step: each connection
693 in it is a hypothesis supplied rather than a result derived, and Table 1 names
694 all eight. What follows takes those conditions in turn, as commitments the
695 proposal incurs, as measurements that would bear on them, and as questions
696 the framework leaves open.

697 8.1 Unity and a minimal self as empirical commit- 698 ments

699 We propose the state glued from compatible local descriptions as a corre-
700 late of conscious unity, and its accurate internal reconstruction as a correlate
701 of a minimal self. The motivations are that different contents belong to

one experienced situation and that the situation is experienced from a perspective. The conditional results identify relations a realization would need: sufficient installed resources for the specified phase model, agreement about shared content, and an encoding that supports reconstruction at the required fidelity.

Compatibility allows different descriptions to contribute different information. It also permits uncertainty and conflicting judgments: apparent cup position and judged cup position can be distinct quantities in a consistent description of their discrepancy. Which quantities are shared must be determined independently. Choosing them solely to secure agreement would deprive the proposal of empirical force.

The minimal-self proposal concerns the perceiver's position and condition within the situation, such as the cup's location relative to a body that can reach it. The encoding could be implicit and need carry no autobiographical narrative. Its proposed necessity is philosophical: even an unreflective episode has a perspective. Gluing by itself requires no self-encoding, and reconstructability alone supplies no functional role. Altering the encoded hand or viewpoint while holding the external scene fixed should change estimates of the perceiver's relation to the cup as the readout predicts. Whether that represented perspective is experienced remains a further question.

722 8.2 What a test of the full proposal must establish

723 Both uniqueness results concern specified objects. They choose neither the
724 cover nor the encoding, and do not select between candidate subjects. In-
725 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
726 here selection requires independent evidence about neural organization. A
727 constant readout can contract and fix a predetermined state while discard-
728 ing its input. The family criterion in Section 4.2 tests the sensitivity that
729 this construction lacks, provided the relevant states and fidelity are declared
730 independently.

731 A test must separately measure coupling relative to diffusion, content
732 compatibility and self-reconstruction, alongside an independent endpoint for
733 experience. Meeting all three without a unified episode would challenge suffi-
734 ciency. A unified episode with incompatible descriptions would challenge the
735 unity identification; one lacking the specified self-encoding would challenge
736 the reflexive requirement. Awakening, dreaming and dissociative states offer
737 possible settings, but absence of report cannot by itself define absence of
738 experience.

739 The next physical requirement is a calibrated geometry–coupling relation
740 and resolved phase dynamics. The installation bound contributes a nec-
741 essary resource condition where its hardware representation applies; cortical
742 mode prices and field-mediated coupling require measurement. For the repre-
743 sentational connection, an informative experiment asks whether a controlled
744 change in coordination alters independently decoded content agreement, with

controls for sensory information and decoder quality. The constructed-data estimator establishes feasibility for specified contents. Neural local descriptions, shared overlap encoders and the self-encoding have yet to be identified. The population-size dependence of the phase-to-content bound further limits its quantitative use.

The separation results explain why these measurements must be independent. A stationary distribution leaves several recovery mechanisms indistinguishable. Equal phase order can accompany incompatible descriptions or different task information. The tested plasticity rule preserves coherence while losing its designated template. Each finding concerns the specified model and observable; their role is to delimit the evidence needed for the proposed connections.

8.3 GPUs and the limits of no-go claims

For a large language model (LLM) running on graphics processing units (GPUs), the framework requires identifying a physical system whose local descriptions form one state and whose internal encoding reconstructs that same state. The coupling-gated route E56–E78 requires a calibrated physical coupling, its noise regime and a connection from coordination to content agreement. An attention matrix or a shared clock supplies none of these identifications on its own. Computing a field model likewise leaves its realization in the computing substrate to be established. These physical obligations also apply to the cortical proposal.

767 Unity requires the candidate’s local descriptions to agree on indepen-
768 dently identified shared quantities (Section 3). Common timing and fluent
769 output do not establish that agreement. Digital representations can sat-
770 isfy compatibility, however, so finite hardware is compatible with the gluing
771 construction. The empirical commitment is to identify the relevant cover
772 and contents with one experienced situation; mathematical uniqueness alone
773 selects no subject.

774 For self, the system must carry an encoding $E(s)$ and an internal readout
775 R of the same global state. E89 requires $F = R \circ E$ to contract and fix
776 the state produced by gluing. A constant readout shows why a unique fixed
777 point alone is insufficient for informative self-representation. The additional
778 family criterion in Section 4.2 supplies a test: if two relevant macrostates
779 satisfy $E(s) = E(t)$ and $d(s, t) > 2\varepsilon$, Eq. (10) rules out reconstructing both
780 within error ε . An intervention outside the encoding region’s causal past
781 gives a corresponding obstruction before a declared communication deadline.
782 Identical verbal reports alone do not establish identical internal codes. These
783 conditional no-go results exclude a specified reconstruction at a specified
784 fidelity or deadline. They apply to biological and digital candidates alike.

785 The hardware comparisons in Supplemental Material have similarly re-
786 stricted scope. Fixed wiring can impose a correlation disadvantage, and
787 point-supported kernels vanish under an atomless continuum measure. Nei-
788 ther comparison makes the representation conditions unattainable on GPUs.
789 A universal exclusion would need a necessary condition for consciousness and

790 a proof that every relevant implementation violates it. The present condi-
791 tional composition supplies neither.

792 Whether experience requires a living substrate is the explicit subject of
793 biological naturalism (Seth, 2026a,b). On that view the properties ground-
794 ing experience are those of self-sustaining organisms, and simulating them
795 does not secure them. The conditions stated here name no biological mecha-
796 nism. Computation described as such does not satisfy them either: a candi-
797 date must exhibit a physically realized coupling in a measured noise regime,
798 agreement about independently identified shared quantities, and an internal
799 reconstruction meeting a declared fidelity. Whether a realization of those
800 conditions that is not alive would be accompanied by experience is the point
801 at issue, and these results do not settle it. A related position locates the
802 difference in the mode of computation rather than in life as such: biologi-
803 cal systems compute across scales that do not separate, and alongside dis-
804 crete operations carry continuous-valued dynamics belonging to their phys-
805 ical medium (Milinkovic & Aru, 2026). The coupling condition stated here
806 is one measurable instance of that requirement, fixed to a physical field in a
807 declared noise regime, and a description of such a field does not satisfy it.

808 The neuroscientific case against *human-like* consciousness in text-only
809 LLMs concerns embodied sensory interaction, thalamocortical organization
810 and biological regulation linking perception and action to survival (Aru et
811 al., 2023). Their absence supports skepticism, conditional on their causal
812 relevance to experience. Those mechanisms are not requirements of the ab-

813 stract gluing and reconstruction definitions. Workspace-like representations
814 reported in some LLMs (Gurnee et al., 2026) provide candidate organization
815 to investigate. Within this framework, the relevant evidence would identify
816 compatible contents and an informative internal reconstruction, or demon-
817 strate their failure under controlled interventions.

818 **8.4 Explanatory scope**

819 Integrated-information, global-workspace and predictive-processing accounts
820 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
821 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
822 candidate physical interaction and conditional relations between local content
823 and self-encoding. Gluing and reconstruction permit multiple substrates;
824 the electromagnetic commitment enters through physical calibration. The
825 formalization checks implications under their hypotheses, and its witnesses
826 establish realizability on declared models. They do not together supply a
827 cortical agent.

828 The explanatory question is whether the specified relations occur in neural
829 activity and matter for experience. In the cup example, this means establish-
830 ing how coordination helps maintain one represented situation and how the
831 perceiver’s place within it contributes to the episode. The proposal identi-
832 fies candidate correlates and tests of that identification. Why those relations
833 would be accompanied by experience remains unexplained.

834 **Data and code availability**

835 The Lean development, simulation scripts, tests and saved numerical sum-
836 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
837 `cortical_coherence`. Publication simulation values are generated from
838 saved summaries without rerunning production sweeps. The exploratory
839 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-
840 processing and selection are described in Supplemental Material. The repos-
841 itory contains the reproducible analysis and does not redistribute the raw
842 recordings.

843 **Funding and competing interests**

844 This research received no external funding. The author declares no compet-
845 ing interests.

846 **Use of AI tools**

847 AI assistance to this work was substantial and is not separable by task. Sev-
848 eral coding assistants and chat interfaces were used in an interleaved way
849 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
850 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
851 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
852 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,

853 accessed with the Hermes CLI. Between them they contributed to the Lean
854 development, the simulation and analysis code, and the drafting and revision
855 of both documents. Because the tools were used in combination and in alter-
856 nation, no section, proof, figure or result is attributable to any one of them,
857 and none is claimed to be. The author is responsible for all content, including
858 everything a tool produced. Machine checking establishes the stated Lean
859 results under their hypotheses; it does not validate biological interpretations
860 or replace scientific review of generated code and prose.

861 **References**

- 862 Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable
863 electric fields emerge from and allow representational drift. *NeuroImage*,
864 253, 119058.
- 865 Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated informa-
866 tion theory: from consciousness to its physical substrate. *Nature Reviews*
867 *Neuroscience*, 17(7), 450-461.
- 868 Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the
869 Mind. *Oxford University Press*.
- 870 Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a compu-
871 tational theory of conscious processing. *Current Opinion in Neurobiology*,
872 25, 76-84.

- 873 Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature*
874 *Reviews Neuroscience*, 11(2), 127-138.
- 875 Milinkovic, B., & Aru, J. (2026). On biological and artificial consciousness:
876 A case for biological computationalism. *Neuroscience & Biobehavioral Re-*
877 *views*, 181, 106524. doi:10.1016/j.neubiorev.2025.106524.
- 878 Aru, J., Larkum, M. E., & Shine, J. M. (2023). The feasibility of artificial
879 consciousness through the lens of neuroscience. *Trends in Neurosciences*,
880 46(12), 1008–1017. doi:10.1016/j.tins.2023.09.009.
- 881 Seth, A. K. (2026). Conscious artificial intelligence and biological naturalism.
882 *Behavioral and Brain Sciences*, 49, e315. doi:10.1017/S0140525X25000032.
- 883 Seth, A. K. (2026). The stuff matters: Consciousness, compu-
884 tation, and biology. *Behavioral and Brain Sciences*, 49, e366.
885 doi:10.1017/S0140525X26106906.
- 886 Gurnee, W., et al. (2026). Verbalizable Representations Form a
887 Global Workspace in Language Models. *Transformer Circuits Thread*.
888 arXiv:2607.15495.
- 889 Aguilera, M., Millidge, B., Tschantz, A., & Buckley, C. L. (2022). How par-
890 ticular is the physics of the free energy principle?. *Physics of Life Reviews*,
891 40, 24-50.
- 892 Biehl, M., Pollock, F. A., & Kanai, R. (2021). A technical critique of some
893 parts of the free energy principle. *Entropy*, 23(3), 293.

894 Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Em-
895 peror's New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.

896 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
897 memory network formation. *Cerebral Cortex*, 33(17), 9877-9895.

898 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
899 Electric fields sculpt neural activity and "tune" the brain's infrastructure.
900 *Progress in Neurobiology*, 226, 102465.

901 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
902 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
903 9(8), 45-60.

904 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
905 testable hypothesis about the characteristics of conscious as opposed to
906 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

907 McFadden, J. (2020). Integrating information in the brain's EM field: the
908 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
909 niaa016.

910 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
911 guide neocortical network activity. *Neuron*, 67(1), 129-143.

912 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
913 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
914 95-103.

- 915 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
 916 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.
- 917 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
 918 on brain circuits in rats and humans. *Nature Communications*, 9, 483.
- 919 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
 920 *logical Reviews*, 98(1), 239-389.
- 921 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
 922 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
 923 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.
- 924 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
 925 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
 926 ity of complexity measures to capture dreaming during propofol sedation.
 927 *Scientific Reports*, 15, 32746.
- 928 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
 929 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
 930 brain. *Science*, 342(6156), 373-377.
- 931 Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space.
 932 *Physiological Reviews*, 88(4), 1277-1340.
- 933 Barbour, B. (2017). Analysis of claims that the brain extracellular impedance
 934 is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.

935 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
936 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
937 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.

938 Landauer, R. (1961). Irreversibility and heat generation in the computing
939 process. *IBM Journal of Research and Development*, 5(3), 183-191.

940 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
941 *of Physics A: Mathematical and General*, 9(8), 1387-1398.

942 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
943 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.

944 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
945 of prediction. *Physical Review Letters*, 109(12), 120604.

946 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
947 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

948 Horowitz, J. M., & Esposito, M. (2014). Thermodynamics with continuous
949 information flow. *Physical Review X*, 4, 031015. doi:10.1103/PhysRevX.4.
950 031015.

951 Ito, S., & Sagawa, T. (2013). Information thermodynamics on
952 causal networks. *Physical Review Letters*, 111, 180603. doi:10.1103/
953 PhysRevLett.111.180603.

- 954 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
955 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.
- 956 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
957 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
958 181.
- 959 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
960 archive.pitt.edu/115.
- 961 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
962 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
963 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.
- 964 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
965 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
966 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 967 Berglund, N., & Gentz, B. (2002). Pathwise description of dynamic pitchfork
968 bifurcations with additive noise. *Probability Theory and Related Fields*,
969 122, 341–388. doi:10.1007/s004400100174.
- 970 Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure
971 of non-locality and contextuality. *New Journal of Physics*, 13, 113036.
972 doi:10.1088/1367-2630/13/11/113036.
- 973 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
974 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave

975 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
976 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.

977 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
978 terns underlie complex brain dynamics and are related to cognitive pro-
979 cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
980 023-01626-5.

981 Nicoletti, G., Di Terlizzi, I., & Busiello, D. M. (2026). Balancing information
982 and dissipation with partially observed fluctuating signals. *Physical Review*
983 *Letters*, 137(12), 127101. doi:10.1103/8gx2-rbns.

984 Kiyooka, D., Oomoto, I., Kitazono, J., Saito, Y., Kobayashi, M., Matsub-
985 ara, C., Kobayashi, K., Murayama, M., & Oizumi, M. (2026). Single-
986 cell resolution functional networks during unconsciousness are segre-
987 gated into spatially intermixed modules. *Cell Reports*, 45(2), 116902.
988 doi:10.1016/j.celrep.2025.116902.

989 Oomoto, I., Kiyooka, D., Oizumi, M., & Murayama, M. (2026). Multi-area
990 single-cell calcium imaging dataset of the mouse cortex across wakefulness,
991 sleep, and anesthesia. *Scientific Data*. doi:10.1038/s41597-026-08202-2.